

Tinkercad Design and Revision

Piter Garcia

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Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 5
- **Grade Hub:** Notes [`../grade-5-Notes.md`]
- **Schedule Reference:** `Spring2026_TeachingSchedule_Derek.md` [`../school/Spring2026_TeachingSchedule_I`]
- **Workbook Sheet:** 5th in `Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx`
- **Lesson Focus:** Students build 3D models in Tinkercad by placing shapes, using workplanes, grouping objects, and revising designs through iterative testing.
- **CS Alignment Strength:** Moderate CS / computational artifact design
- **LaTeX Source:** `tinkercad-design-and-revision.tex` [`../tinkercad-design-and-revision.tex`]
- **Bib File:** `tinkercad-design-and-revision.bib` [`../tinkercad-design-and-revision.bib`]
- **Artifacts Folder:** `artifacts/` [`../artifacts/`]

Lesson Identity

- **Likely Class Blocks:** Day 2 – 10:45-11:35 – Pickett, Day 3 – 10:45-11:35 – McGlashon, Day 5 – 10:45-11:35 – Pecora
- **Main Lesson Family:** Tinkercad Design and Revision
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- navigating a digital design workspace
- breaking a design into component shapes
- using grouping and holes to refine an artifact
- revising a computational artifact after testing or feedback

CS Standards Alignment

- **1B-AP-11:** Decompose problems into smaller, manageable subproblems. Why it fits here: Students break a complex object into shapes and grouped steps.
- **1B-AP-13:** Use an iterative process to plan the development of a computational artifact. Why it fits here: Students revise, regroup, and refine the design after testing how it looks or works.
- **1B-IC-20:** Seek diverse perspectives for the purpose of improving computational artifacts. Why it fits here: Students benefit from feedback on whether the artifact is understandable and functional.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 5 design sequence on the workbook sheet through digital modeling and revision.
- Supports New York-facing technology and design outcomes by requiring students to create and iteratively improve a digital artifact.
- This packet is less programming-heavy but still part of computing because students use structured digital design processes and iterative computational artifact creation.

Lesson Content and Concepts

- computational artifact
- decomposition
- grouping
- iteration
- spatial planning

Evidence / Proof of Instruction

- **Transcript-derived note:** 260304 Grade 5 Pickett Tinkercad [../transcript-derived-lessons/260304-grade-5-pickett-tinkercad.md] The note explicitly lists shape placement, planes, grouping, and hole creation as taught content.

- **Source transcript:** 260304 Grade 5 Pickett Tinkercad [../transcripts/260304-grade-5-pickett-tinkercad.md] The source transcript documents classroom guidance around how to enter the workspace and manipulate design tools.
- **Observed teacher moves:** Instruction is evidenced by modeling the design workspace, sequencing tool use, and revising models rather than leaving students to explore randomly.
- **Likely student proof:** Saved Tinkercad models, screenshots of revisions, and student explanations of grouping or hole use count as proof.

Assessment Evidence

- Student can place and manipulate at least one basic shape intentionally.
- Student can use grouping or holes to improve a design.
- Student can explain one revision they made and why it improved the artifact.

UDL and Inclusion Supports

Multiple Representation

- Project the workspace and name only the tools needed for the current task.
- Use side-by-side before/after screenshots to show revision value.
- Keep navigation vocabulary tied to visible actions.

Multiple Action / Expression

- Allow students to sketch first, then build.
- Provide a starter file or partially built model for students who need lower entry load.
- Accept oral explanation of design choices in addition to written reflection.

Multiple Engagement

- Use real object examples so the design has a concrete target.
- Frame revision as improvement, not mistake-fixing.
- Let students compare two design choices and defend one.

How We Are Already Making It Inclusive

- Transcript evidence shows the lesson was already focused on tool-specific instruction, not unstructured exploration.
- The workspace lends itself to strong visual modeling, which supports many learners.
- Iterative revision naturally creates opportunities for students to succeed through multiple attempts.

How to Improve Inclusion Next Time

- Add a printed icon guide for move, rotate, group, hole, and workplane.
- Capture before/after screenshots in **artifacts/** to prove revision.
- Create a two-level challenge: basic shape composition and advanced grouped design.
- Add a quick oral checkpoint where students explain their design plan before building.

Materials and Setup

- Student devices
- Tinkercad accounts or access
- Projected workspace
- Reference objects or sample images

Teacher Moves / Script / Routines

- Model one object by decomposing it into shapes before touching the software tools.
- Pause after the first shape placement to make sure all students are in the same view.
- Show grouping and hole tools as solutions to a design problem, not as disconnected features.
- Close with a revision share-out.

Common Errors and Debugging

- Students may get lost in navigation and view control before understanding design concepts.
- Grouping too early can make editing harder.
- Students may overbuild without checking whether the design still matches the goal.

Extensions and Simplifications

- Extension: redesign the object for a different user need or constraint.
- Simplification: provide a three-shape target with fixed measurements.
- Extension: compare two design plans and choose the more efficient build process.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References